

Advancing Our Understanding of Psychological Flow: A Scoping Review of Conceptualizations, Measurements, and Applications

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Research on psychological flow is well established, although criticisms remain regarding conceptual and measurement issues associated with the construct. This scoping review maps flow-related research across scientific disciplines, examining the conceptualization, measurement instruments, and outcomes of flow between 2012 and 2019. Across 236 sources that met the review criteria, 108 different flow-related constructs were measured by 141 instruments, and 84 possible antecedents were identified. Despite the varied approaches, a common set of overarching antecedent constructs included “optimal challenge” and “high motivation,” and recurring characteristics of the flow experience itself included “absorption,” “effort-less control,” and “intrinsic reward.” Applied studies—albeit inconsistent in approach and largely correlational in nature—predominantly linked flow to “positive development” (i.e., well-being and health), “high functioning,” and “further engagement.” We contextualize the findings of the review relative to important work on flow that has recently emerged (following the review period)—in doing so, we hope this review offers a contemporary framework that can be used for the study of flow across scientific disciplines.

Public Significance Statement

This review scopes flow research across all scientific disciplines and identifies key commonalities and differences in how researchers have conceptualized the psychological state of flow—an optimal experience often referred to as being “in the zone.” It addresses inconsistencies and concerns of practical application that characterize some contemporary flow-related models and studies. This review also provides a framework—derived from the findings of the review—for the study of flow antecedents and experiential dimensions.

Keywords: flow theory, performance, stress, motivation, health

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In his pursuit to understand what brings meaning to one’s life, Csikszentmihalyi (1975) described a unique type of intrinsically rewarding, or “autotelic,” experience, which later he defined as “flow.” Deriving from the Greek words *auto* (self) and *telos* (goal), an autotelic experience referred to an experience so rewarding that participation alone was the goal. In Csikszentmihalyi’s original interviews, participants consistently described an intrinsically rewarding, harmonious psychological state of total focus and absorption that facilitated a fluidity to their actions and a playfulness to their efforts as if the experience was “flowing” like water

(Csikszentmihalyi, 1975). Csikszentmihalyi has since outlined nine dimensions associated with the flow experience: (a) challenge-skills balance (i.e., matching perceived skills and challenges in a particular situation); (b) merging of action and awareness (i.e., where deep involvement leads to automaticity and spontaneity); (c) clear goals (i.e., clear purpose and grasp of what to do next); (d) unambiguous feedback (i.e., direct and immediate feedback about one’s effectiveness in the moment); (e) concentration on the task at hand (i.e., exclusion of any unnecessary distractions); (f) sense of control (i.e., a sense of personal control or agency over the activity); (g) loss of self-consciousness (i.e., when the self disappears, worrying about oneself and what others may think of him or herself cease to exist); (h) transformation of time (i.e., one’s subjective experience of time is altered); and (i) autotelic experience (i.e., an intrinsically rewarding activity, an end in itself; adapted from Jackson & Csikszentmihalyi, 1999).

Further work has been undertaken to position these factors as antecedents, core experiences, or outcomes of flow. For instance, Nakamura and Csikszentmihalyi (2009, 2014) outlined that

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“challenge-skills balance,” “clear goals,” and “unambiguous feedback” dimensions may be considered structural task characteristics or proximal conditions to flow, whereas the other six dimensions describe the experience itself. Despite some support for the nine-dimensional model (Jackson & Eklund, 2002; Nakamura & Csikszentmihalyi, 2002), evidence of correlations between the other eight dimensions and the “challenge-skill balance” dimension (Beard & Hoy, 2010; Engeser & Rheinberg, 2008; Jackson & Marsh, 1996) has resulted in many researchers utilizing only the “challenge-skill balance” dimension when attempting to operationalize flow—therefore examining flow as a unidimensional construct (Fong et al., 2015).

Understanding, measuring, and describing flow is by no means straightforward. Flow is complex in nature—it encompasses, for example, both hedonic (i.e., intrinsic reward or pleasure) and Eudemonic (i.e., use of an individual’s skills and abilities leading to individual and cultural progressive development) characteristics (Seligman, 2002; Vittersø & Søholt, 2011). It could also be argued that there is something paradoxical about the high concentration levels (and associated physiological symptoms of effort) that are proposed to co-occur alongside effortless attention and a fluidity of performance. Despite such complexities, flow research has become and remains popular across sport, educational, health, and organizational psychology domains. Importantly though, alongside empirical reports of flow antecedents, correlates, experiences, and interventions, work in these areas over the last decade has also been characterized by critiques of the conceptualization and measurement of the construct. Broadly, these critiques have highlighted four important considerations—specifically, issues relating to construct validity, theoretical compatibility, relational ambiguity, and definitional inconsistency.

With respect to *construct validity*, it has been recognized that flow dimensions may vary between activities—that is, that not all nine dimensions are necessary to experience flow in varying contexts (e.g., Boyd et al., 2018; Corbato, 2018; Kawabata & Mallett, 2016). For example, in an educational setting, Heutte et al.’s (2016) factor analysis showed the dimensional use of “cognitive absorption,” “time transformation,” “loss of self-consciousness,” and “autotelic experience” to measure flow (dimensions)—challenge/skill balance, clear goal, unambiguous feedback and high sense of control—were merged into one antecedent (factor). Further, Sinnamon et al. (2012) suggested that the dimensions of “merging of action and awareness” and “loss of self-consciousness” may not be valid dimensions for musicians, and Swann et al.’s (2012) systematic review in elite athletes highlighted that less than 30% of athletes in included studies reported the dimensions of “loss of self-consciousness” and “transformation of time.” In addition to contextual variability, construct validity has also been questioned due to (a) some researchers not including any of the nine dimensions when measuring flow (for example, Zito et al., 2018); (b) certain dimensions (for example, “time transformation”) exhibiting only modest relationships with the other dimensions (See, for example, see Jackson & Eklund, 2002); and (c) additional experiential dimensions beyond the nine “core” aspects being frequently reported (for example, “bodily sensations”; Chavez, 2008; Jackman et al., 2017), including other widely used psychological constructs (for example, arousal, motivation, confidence; see Swann et al., 2012) that do not naturally fit into one of the nine dimensions.

In terms of *theoretical compatibility*, Csikszentmihalyi’s (1990) nine-dimensional model utilizes terms that are not commonly utilized in scientific research (for example, autotelic experience, merging of action and awareness), and in some respects, these constructs can be difficult to align with existing research or theory. For example, self-efficacy appears to be characterized in three different flow dimensions—“challenge-skill balance,” “clear goals,” and a “sense of control” (for example, Jackson & Csikszentmihalyi, 1999)—which indicates a degree of intradimensional overlap and poses problems for any theoretical integration efforts. Moreover, Swann et al. (2018) suggested that Csikszentmihalyi’s (1990) nine-dimensional model should not be considered a “theory” as it lacks clear causal constructs and outcomes, making it difficult to examine experimentally or integrate with other theories.

Regarding *relational ambiguity*, Engeser and Schiepe-Tiska (2012) argued that although the nine dimensions of flow actually include antecedents, experiential constructs, and outcomes to flow, no interactional relationships are proposed or offered in the model (see also Guertin, 2016). Consequently, for example, some researchers regard concentration and enjoyment as elements of the flow experience (for example, Ghani, 1995), while others view concentration as an antecedent and enjoyment as an outcome of flow (for example, Chen, 2006). Finally, in terms of *definitional inconsistency*, it is recognized that definitions of flow vary immensely—even those presented by Csikszentmihalyi himself—and that many are not wholly compatible with the nine-dimensional model (Cheng, 2014; Hoffman & Novak, 2009; Swann et al., 2018). To illustrate, working definitions of flow include:

- “Flow is a positive valenced state (affective component), resulting from an activity that has been appraised as an optimal challenge (cognitive component), characterized by optimized physiological activation (physiological component) for full concentration on coping with environmental/task demands (behavioral component)” (Peifer, 2012, p. 149).
- Flow incorporates the extent to which (a) the user perceives a sense of control over the computer interaction, (b) the user perceives that his or her attention is focused on the interaction, (c) the user’s curiosity is aroused during the interaction, and (d) the user finds the interaction intrinsically interesting” (Webster et al., 1993, p. 413).
- “The state occurring during network navigation which is (1) characterized by a seamless sequence of responses facilitated by machine interactivity, (2) intrinsically enjoyable, (3) accompanied by a loss of self-consciousness, and (4) self-reinforcing” (Novak & Hoffman, 1997, p. 2).
- “The holistic sensation that people feel when they act with total involvement” (Csikszentmihalyi, 1975, p. 36).

These conceptual inconsistencies and (as some have argued) ambiguities have caused difficulties for researchers and practitioners when attempting to compare and synthesize flow research (Auld, 2014). Mahnke et al.’s (2012) flow review in information systems, for example, indicated that 75% of the authors of studies included in their review either modified the original constructs of flow (beyond adaptation to the research context), combined constructs from different concepts (for example, Huang, 2006; Novak et al., 2000; Shin, 2006), or customized terminology and interpretation of constructs to better suit their own context (for example,

cognitive absorption in education; Heutte et al., 2016)—highlighting the need for an overarching definition. Further, Swann et al. (2018) contended that the current (that is, most widely accepted and used nine-dimensional) flow model is too descriptive and not sufficiently mechanistic, and argued as a result that flow science may be at a “crisis point.” More specifically, these authors also asserted that the flow model cannot be considered a “theory” as it is missing specific definitions and propositions of how interrelated constructs create clear causal relationships for a particular phenomenological occurrence (or nonoccurrence).

Despite the substantial number of flow-related studies conducted in the past decade, only four systematic reviews of psychological flow literature have been published—one with an emphasis on musical applications (Chirico et al., 2015), one on game-based learning (Perttula et al., 2017), one in computer-mediated learning environments (Auld, 2014), and one on elite athleticism (Swann et al., 2012). Although informative, these reviews have typically focused on a specific domain or specific questions, rather than examining flow science in its entirety (that is, across multiple conceptual issues and multiple contexts). Equally, informative books, chapter narratives, and conceptual reviews have helped to synthesize aspects of flow research (for example, Abuhamdeh, 2020; Engeser, 2012a; Harmat et al., 2016; Nakamura & Csikszentmihalyi, 2014), including the collected works of Mihaly Csikszentmihalyi (2014a); however, no formal review has been undertaken to chart psychological flow research across domains (for example, sport, education) and scientific disciplines (for example, physiology and neurobiology).

The broad purpose of this study, therefore, was to undertake and present a comprehensive scoping review that brought together—and charted, in one place—the varied conceptualizations of flow. Specifically, our aim in this review was to describe the consistency (and inconsistency) in constructs, dimensions, and working definitions utilized to describe flow, and to chart the varied antecedents, measurement instruments, and outcomes utilized when operationalizing flow. In line with recommendations for cross-disciplinary approaches, and calls to understand flow phenomenologically, cognitively, neurologically, physiologically, socially, and behaviorally (Beard, 2015; Csikszentmihalyi, 2014b; Kawabata & Mallett, 2016; Marr, 2001; Peifer, 2012; Rich, 2013; Stoll & Pithan, 2016), a secondary aim (if possible) was to highlight any common findings that could be used and operationalized across scientific disciplines.

Given the aims outlined above, we felt that a narrow search strategy using predetermined parameters and set domains (that is, a PRISMA-adherent systematic review: Moher et al., 2009) may limit the scope of our review. Our broad goal, however, did align closely with Arksey and O'Malley's (2005) and Tricco et al.'s (2016) guidelines for scoping reviews (that is, examining the extent, range, and nature of research activity; determining the value of undertaking a systematic review; summarizing and disseminating research findings; and identifying research gaps in the existing literature). As a result, a scoping review methodology was chosen for the study. The scoping review adhered to The Joanna Briggs Institute (JBI) guidelines (Peters et al., 2015) and the PRISMA extension for scoping reviews (PRISMA-ScR; Tricco et al., 2018), and was registered a priori with JBI on August 1, 2019 to ensure methodological rigor (Colquhoun et al., 2014). To adhere to scoping review procedures (Peters et al., 2015), the

objectives, inclusion criteria, methods of analysis, and a priori search terms were specified in advance and documented in a protocol.

Method

Inclusion/Exclusion Criteria

There were no exclusion criteria for methodological approaches, participant characteristics, geography, or language. Literature from 2012 through to 2019 (the year at commencement of the review) that specifically analyzed “flow” conceptually was included to ensure theoretical rigor. The year 2012 was selected due to Engeser's (2012b) seminal theoretical update on flow within “Advances in Flow Research” (Engeser, 2012a)—a text that also included updates and commentaries from numerous flow researchers (including Csikszentmihalyi). Additionally, we selected 2012 because this year was when Peifer's (2012) article was published on the psychophysiological correlates of flow. This work was in part responsible for a subsequent proliferation in what might be termed “nonpsychological” (for example, physiological, psychophysiological, and neuroscientific research) flow-related research (for example, Harris et al., 2017). Finally, 2012 was also the year in which Baumann (2012) presented three different flow definitions (not descriptions) encompassing different research disciplines (general, cognitive, affective). Reviews of the flow literature have been published since 2012 (for example, Abuhamdeh, 2020; Chirico et al., 2015; Nakamura & Csikszentmihalyi, 2014); however, in comparison with those sources, our inclusion criteria were designed to offer broader coverage of flow conceptualization and measurement across multiple disciplines.

Search Strategy

A three-step search strategy was utilized (Peters et al., 2015). Initially, the keyword “flow” was used to begin an iterative search in Embase (1947) and PsycINFO (1806) to analyze common affiliated terms. In Step 2, the selected terms, and negative keywords, were utilized in title and abstract searches within the databases Web of Science (1900), PubMed (1997), Medline (1966), Scopus (1966), Embase (1947), PsycINFO (1806), SPORTDiscus (1930), Google Scholar, OSF, and PsyArXiv. Subsequently, a wider search, (“flow” in the title plus “optimal experience” in the article text) was carried out to ensure all relevant articles were captured and safeguard against any articles that might have been missed from previous, more reductive searches. In Step 3, the keyword “flow” was searched in additional related journals, and the reference list of all identified relevant reports and articles were searched for additional studies. (An online supplemental materials lists the specific terms, negative keyword, and journals.) Where necessary, study authors were contacted for access and clarification.

Data Charting

Search results were recorded in Microsoft Excel, Evernote, Zotero, and Bubbl. Data were charted into the following extraction categories: Document type (that is, book, journal), year of publication, author(s), study location (that is, location of associated institution), scientific discipline, scientific discipline - subdomain,

study population, methodology (experimental work was categorized using Campbell and Stanley's (1963) pre-experimental, quasi-experimental and experimental categories), definition of flow, described constructs, compounding factors, measures, antecedents, intervention, linked outcomes, reported outcomes, quality rating, individual or group orientation, and key findings. Methodological quality was assessed using Kmet et al.'s (2004) standard quality assessment criteria for primary quantitative and qualitative studies; literature reviews were assessed using Oxman and Guyatt's (1988) guidelines, and systematic reviews were assessed using AMSTAR (Shea et al., 2007)—all common assessment tools for reviews. A preliminary meeting was held between all authors to ensure a consistent interpretation of descriptors within the quality assessment tools. The lead author then recorded quality scores, and this recording was verified by the other two authors for accuracy (with discussion in cases where any disagreements arose). Data extraction followed the PRISMA-ScR (Tricco et al., 2018) and JBI scoping review guidelines (Peters et al., 2015).

To ensure high interrater agreement during data extraction (which followed the PRISMA and JBI scoping review guidelines), sample data were extracted by the lead author and cross-checked by the two coauthors for reliability (Brooker et al., 2015) to test the process and ensure accuracy of data extraction. Data from each source were then mined into a predefined data sheet by the lead author. Additionally, longer narrative and sentence extraction was recorded from each source within bubbl.us—a live online mapping tool designed to help group large amounts of data into themes and ensure original data accuracy. Both sets of data were continuously cross-checked by (a) additional researchers acting as “critical friends” (to aid integrity and reflexivity; Appleton, 2011), and (b) the two coauthors—for accuracy of extraction, interpretation, and categorization—on a fortnightly basis over a period of several months. Certain core extraction categories (for example, conceptual dimensions) yielded over 300 different individual items. Consequently, a thematic content analysis (TCA) approach was conducted by the authors in order to identify common concepts/themes across the antecedents, experiential dimensions, and outcomes of flow. Anderson's (2007) TCA protocol was adhered to when synthesizing terms, including identifying distinct units, grouping and regrouping similar and dissimilar units, retaining source language where possible, iteratively relabeling categories based on new source data, and rereading source data postanalysis without looking at final categorizations. Any disagreements were resolved through open discussion between the three authors while revisiting source data to ensure integrity of the source's meaning.

Results

The State of Flow Conceptualization, Measurement, and Application

Literature Search

The literature search resulted in 2,633 citations (see Figure 1). After eligibility screening, 2,391 articles were removed. Three articles were added from reference screening, three books were removed because all the book chapters were already listed, and six articles were not accessible even after contacting the authors.

A final 236 records were included in the review (see supplementary material S1, tab Review Data for full list).

Source Characteristics

Eligible sources included 58 books or book chapters and 178 journal articles, the majority of which were published in 2016, 2017, and 2018 (see supplementary material S2, Figure 1). More than half of all the records came from authors affiliated with European institutions ($k = 133$), followed by North America ($k = 66$), Asia ($k = 18$), Australasia ($k = 14$), South America ($k = 3$), and Africa ($k = 2$). Work reported in journal articles predominantly came from the United States ($k = 60$), followed by Germany ($k = 31$), England ($k = 19$), Italy ($k = 17$), Australia ($k = 14$), Spain ($k = 11$), Hungary ($k = 6$), Taiwan ($k = 5$), and the Netherlands ($k = 5$; see S2, Figures 2 and 3). The most prevalent disciplines included psychology ($k = 173$), psychophysiology ($k = 21$), neuroscience ($k = 19$), and computing ($k = 13$; see S2, Figure 4). Research was most prevalent within the subdisciplines of positive psychology ($k = 42$), sport and exercise psychology ($k = 24$), occupational psychology ($k = 18$), educational psychology ($k = 15$), computing psychology ($k = 12$), personality psychology ($k = 9$), neurobiology ($k = 8$), human computer interaction (HCI; $k = 8$), positive psychophysiology ($k = 7$), cognitive psychology ($k = 7$), and cognitive neuroscience ($k = 7$; see S2, Table 1).

Methodology and Quality Assessment

Pre-experimental studies ($k = 80$), literature reviews ($k = 63$), theorizing literature ($k = 27$), qualitative works ($k = 22$), and confirmatory factor analyses ($k = 21$) were the most widely-used research methods (see S2, Figure 5). The literature was given a high ($k = 27$), medium ($k = 84$), and low ($k = 125$) “quality” score; however, the practical value of these scores was limited—as can be the case when conducting scoping reviews (Tricco et al., 2018)—because much of the theoretical analysis was carried out in narrative reviews or expert commentaries (rather than, for example, experimental studies).

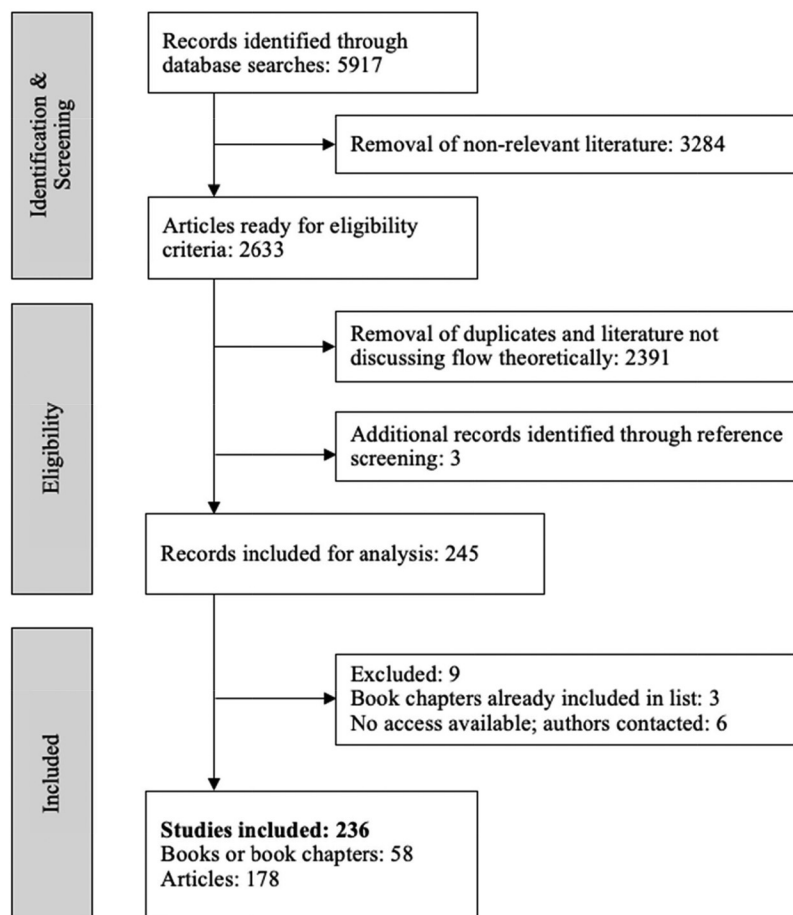
Sample Population

Students ($k = 57$), general population ($k = 24$), athletes ($k = 20$), musicians ($k = 10$), gamers ($k = 10$), workers ($k = 8$), and teachers ($k = 3$) were the samples most utilized (see S2, Table 2). Studies were focused on individuals ($k = 222$), groups ($k = 10$), or both ($k = 5$).

Definitions and Dimensions of Flow

The majority (80%) of authors created their own phrase (that is, flow definition) or cited one of Csikszentmihalyi's statements to describe flow. Only three exact descriptive phrases were utilized by multiple sources (namely, “holistic sensation that people feel when they act with total involvement,” $k = 14$; “being fully absorbed in an activity,” $k = 4$; “in which people are so intensely involved in an activity that nothing else seems to matter; the experience itself is so enjoyable that people will do it even at great cost, for the sheer sake of doing it” $k = 3$). Twenty-seven records (11%) did not list a clear definition or description of the focal construct. In total, 270 psychological and physiological markers, dimensions, or terms were associated with the conceptualization of flow. A total of 150 different constructs, terms, or dimensions were utilized to describe the flow experience itself. Csikszentmihalyi's

Figure 1
PRISMA Flow Diagram Identifying the Included Records (Peters et al., 2015)



nine dimensions were referred to together in 109 (46%) of the records. Of those nine dimensions, the most frequently described were “sense of control” ($k = 166$) and “total concentration/focus” ($k = 158$), followed by “immersion” or “absorption” ($k = 115$), “joy” ($k = 66$), “optimal experience” ($k = 66$), “enjoyment” ($k = 66$), “total engagement” ($k = 66$), “motivation” ($k = 31$), “intrinsically rewarding” ($k = 28$), “nothing else matters” ($k = 25$), “full attention” ($k = 25$), “effortless attention” ($k = 23$), and “effortless involvement” ($k = 23$). See S2, Tables 3 and 4 for a full listing of these terms.

Antecedents

Eligible sources identified 84 different antecedents to flow, described 158 compounding factors, and measured 30 variants in applied studies (see S2, Tables 5 and 6). Csikszentmihalyi’s (Nakamura & Csikszentmihalyi, 2009) three theorized “preconditions” to flow—namely challenge-skill balance (described, $k = 149$ /measured, $k = 52$), clear-goals (described, $k = 105$ /measured, $k = 34$), and unambiguous feedback (described, $k = 102$ /measured, $k = 32$)—were the most widely utilized, and were often reported together ($k = 96$). “Skills-demand fit” was utilized in work from European researchers (e.g., Engeser & Schiepe-Tiska, 2012), while “concentration,” “attention,” “sense of control,” and “positive affect” were frequently described across sources (see S2, Tables 5

and 6). Additionally, “autonomy,” “choice,” “motivational competence,” and “interest” were repeatedly observed as proposed or measured antecedents, indicating that there is a motivational antecedent additional to the challenge-skill balance dimension. In the two applied experimental studies (Colzato et al., 2018; Peifer et al., 2015), challenge and arousal levels were targeted through manipulating the level of task difficulty, vagus nerve stimulation, and cortisol levels (see also S1, tab Applied Studies).

Outcomes

Flow was described to have 133 different outcomes in all studies, while studies targeting outcomes of flow ($k = 75$) measured 45 varying outcomes (see S2, Tables 7 and 8) across a range of designs; 3% experimental, 5% quasi-experimental, and 75% pre-experimental designs—other designs included reviews, confirmatory factor analyses and qualitative approaches. Flow was (or proposed dimensions of flow were) most frequently positively correlated with performance ($k = 6$), behavioral engagement ($k = 4$), learning ($k = 3$), joy ($k = 2$), further flow ($k = 2$), enjoyment ($k = 2$), well-being ($k = 2$), and mental health ($k = 2$). Measured outcomes were spread across emotional and motivational ($k = 18$; 27%), behavioral ($k = 17$; 25%), cognitive ($k = 17$; 25%), physiological ($k = 14$; 21%) and trait-based ($k = 1$; 1%) indices (note:

studies reported more than one outcome). Findings were largely correlational in nature and researchers utilized inconsistent conceptual and measurement approaches; therefore, any causal conclusions are premature. In the few experimental studies, reported outcomes included further flow, high levels of arousal, and inverted U-shape heart-rate-variability (HRV)—though arousal levels and HRV may better represent measurement methods associated with experiential components of flow (rather than outcomes).

Measurement

Overall, questionnaires assessing flow (or dimensions thereof) appeared in 210 sources—including 39 experience sampling method (ESM) studies—followed by electrocardiogram (ECG) measures ($k = 36$), electroencephalography (EEG) measures ($k = 17$), and functional MRI (fMRI) measures ($k = 16$). The most commonly utilized instruments were multicomponent measures within ESM studies ($k = 33$), Flow State Scales ($k = 29$; FSS-2; Jackson & Eklund, 2002), Dispositional Flow Scales ($k = 22$; DFS-2; Jackson & Eklund, 2002), Flow Short Scale ($k = 20$; Engeser & Rheinberg, 2008; Rheinberg et al., 2002), The Work-related Flow inventory ($k = 10$; WOLF; Bakker, 2008), and Swedish Flow Proneness Questionnaire ($k = 9$; SFPQ; Ullén et al., 2012). See S2, Table 9 and 10. In these instruments, scored responses offered insight into flow “intensity” or “proneness” levels, but no instrument clearly identified whether or when a person was “in” flow or not. These instruments also did not account for multiple flow entries (and exits) in a given experience.

In terms of psychological assessments, 108 constructs or concepts were measured through 141 different instruments (see S2, Tables 4 and 9). The nine dimensions were the most frequently assessed constructs, with “level of challenge” ($k = 58$) and “level of skill” ($k = 33$) also being measured independently from “challenge-skill balance” ($k = 107$). In addition, “immersion” or “absorption” ($k = 54$), “positive affect” ($k = 34$), “reasons for participation” ($k = 33$), “satisfaction” ($k = 33$), “enjoyment” ($k = 32$), “fluency” ($k = 25$), and “importance” ($k = 23$) were also commonly measured.

In terms of physiological assessments, sources examined markers of flow associated with the brain ($k = 40$), heart ($k = 38$), facial expressions/movement ($k = 16$), skin ($k = 12$), hormones ($k = 13$), breathing ($k = 7$), and muscle movement ($k = 4$). Physiological and neuroscientific instruments predominantly targeted arousal levels ($k = 66$) and the autonomic nervous system ($k = 4$; for example, heart-rate-variability; HRV), active geographical areas in the brain ($k = 35$; for example, blood flow), valence ($k = 14$; for example, corrugator supercilii muscle movement), attention ($k = 9$; eye pupil movement), affect and reward processing ($k = 7$; ventral striatum or putamen activity), fluency of movement ($k = 4$; for example, cortico-muscular coherence), and mental effort ($k = 3$; for example, heartbeat stroke volume). See S1, tab Measured Terms, for a detailed listing. Arousal levels and HRV (that is, the variability and coherence of the length of inter heart-beat intervals) were targeted to examine optimal levels of challenge, concentration, stress, sense of control, and activity within the autonomic nervous system (ANS). Brain activity was used to measure cognitive and biological functions and associations with attention, self-consciousness, and time monitoring. Valence and affect were targeted to examine the attractiveness of the

experience and the positive or negative affect that flow yielded, as well as indicating the likelihood for subsequent focus. Attention and mental effort were assessed in order to examine high attentional activity and paradoxical reports of effortless attention. Similarly, fluency of movement was examined to understand effortless action, reduced pain, and feeling of smoothness in actions when exerting high levels of effort. Cardiac output and respiratory rates were measured to analyze mental effort and autonomic nervous system (ANS) activity. In summary, preliminary physiological findings reveal a somewhat coherent set of physiological markers that suggest flow may differ from alternative physiological modes of operating.

Physiological findings appeared to confirm an optimal facilitative level of arousal that corresponds with a down regulation of neuroanatomical components of the brain that are associated with effortful activity (Peifer, 2012). For example, higher flow intensity levels (as measured by Flow Short Scale; Rheinberg et al., 2002) have been linked in an inverted u-shaped relationship with sympathetic arousal and cortisol production, while parasympathetic activity has been linearly and positively correlated with higher flow intensity levels during coactivation of sympathetic activity (e.g., Peifer et al., 2014). Further, higher intensities of flow have been characterized by a faster respiratory rate, deeper respiration, moderate heart rate, and moderate HRV, thereby suggesting that flow occurs with highly focused attention that is facilitated through an increased parasympathetic modulation of sympathetic activity (Bian et al., 2016; Tian et al., 2017). A higher sense of control during high levels of flow intensity is physiologically evident from eye movement and pupil dilation; specifically, flow has been linked with higher attention signified by increased pupil size and reduced blinking (for example, Kim et al., 2013). Further, through the measure of functional magnetic resonance perfusion imaging (fMRPI), increases in neural activity in the left anterior inferior frontal gyrus (IFG) and the left putamen suggest a deeper sense of cognitive control and coding of increased outcome probability (see Ulrich et al., 2014). Further, this level of control appears to have a physiological basis of ease, fluidity, or effortlessness as indicated by EMG and EEG reports of high levels of cortico-muscular coherence (that is, synchronization of neural activity between the brain and body) during higher levels of flow intensity (Cheron, 2016). The above physiological findings help to suggest how an individual can be challenged and subsequently highly aroused and alert, yet also able to act calmly and with ease as psychologically evident during flow (see Effortless Control section below for further discussion).

Physiological explanations were also offered for the phenomenological characteristics of absorption, loss of self-consciousness, and time transformation. Studies examining electrical activity in the brain (often measured with EEG) linked states of flow with reduced beta brain wave frequencies, suggesting a reduction in conscious thinking and self-reflexive activity such as self-consciousness (for example, Pluta et al., 2018). Similarly, neuroanatomy studies (often using fMRI to understand which areas of the brain, and associated faculties, are activated and inhibited) revealed that the brain’s attentional resources are highly engaged during high flow intensities, while brain activity associated with worry and self-concern about one’s performance is inhibited (for example, de Sampaio Barros, 2018; Sadlo, 2016). Additionally, researchers have linked flow with a moderate (that is, not high or

low) default mode network (DMN: a large-scale network of brain regions that is responsible for our normal modes of conscious thinking and waking consciousness) activation (for example, Esterman et al., 2013) and transient hypofrontality (for example, Nakamura & Roberts, 2016). Such work physiologically indicates the loss or reduction of normal modes of self-consciousness, time monitoring, and awareness that are often considered to be absent during flow. Lastly, skin conductance tests (that is, testing the ability of the skin to resist electrical currents) have been used to measure an individual's stress, arousal, and valence in order to examine the intensity of flow (e.g., Bian et al., 2016).

Moving Toward Conceptual Clarity and Parsimony

Despite the large number of terms ($n = 270$) utilized within the review sources, many of the terms were used across studies and common themes bringing together multiple terms were apparent. For example, only 15 (described)/16 (measured) terms were utilized in more than 10 sources, and Csikszentmihalyi's original nine dimensions accounted for 60% of all described terms and 65% of all measured terms. Accordingly, we sought to synthesize the 270 terms into coherent (and discrete) antecedents, experiential constructs and outcomes through Anderson's (2007) protocol of TCA. Our efforts to do so yielded two broad antecedent dimensions ("optimal challenge" and "high motivation") and three broad experiential dimensions ("absorption," "effort-less control," "intrinsic reward"), which are presented below and supported throughout by citations to relevant literature and review evidence (see Figures 2 and 3). See also S1, tabs Thematic Grouping of Terms for a full list of synthesized terms. Robust conclusions about the outcomes of flow experiences were limited due to methodological issues associated with research in the area. Only 9% of applied studies on flow-related outcomes included a control group, with the majority of studies characterized by correlational and cross-sectional designs. Additionally, various flow instruments have been utilized to measure possible outcomes of the construct, leading to some ambiguity in the identification of key outcomes. Nevertheless, on the basis of research examined in this review, below we also provide preliminary thematic insight into three outcomes of flow: positive development, high functioning, and further engagement. See S1, tab, Thematic Groupings of Outcomes for the full synthesis.

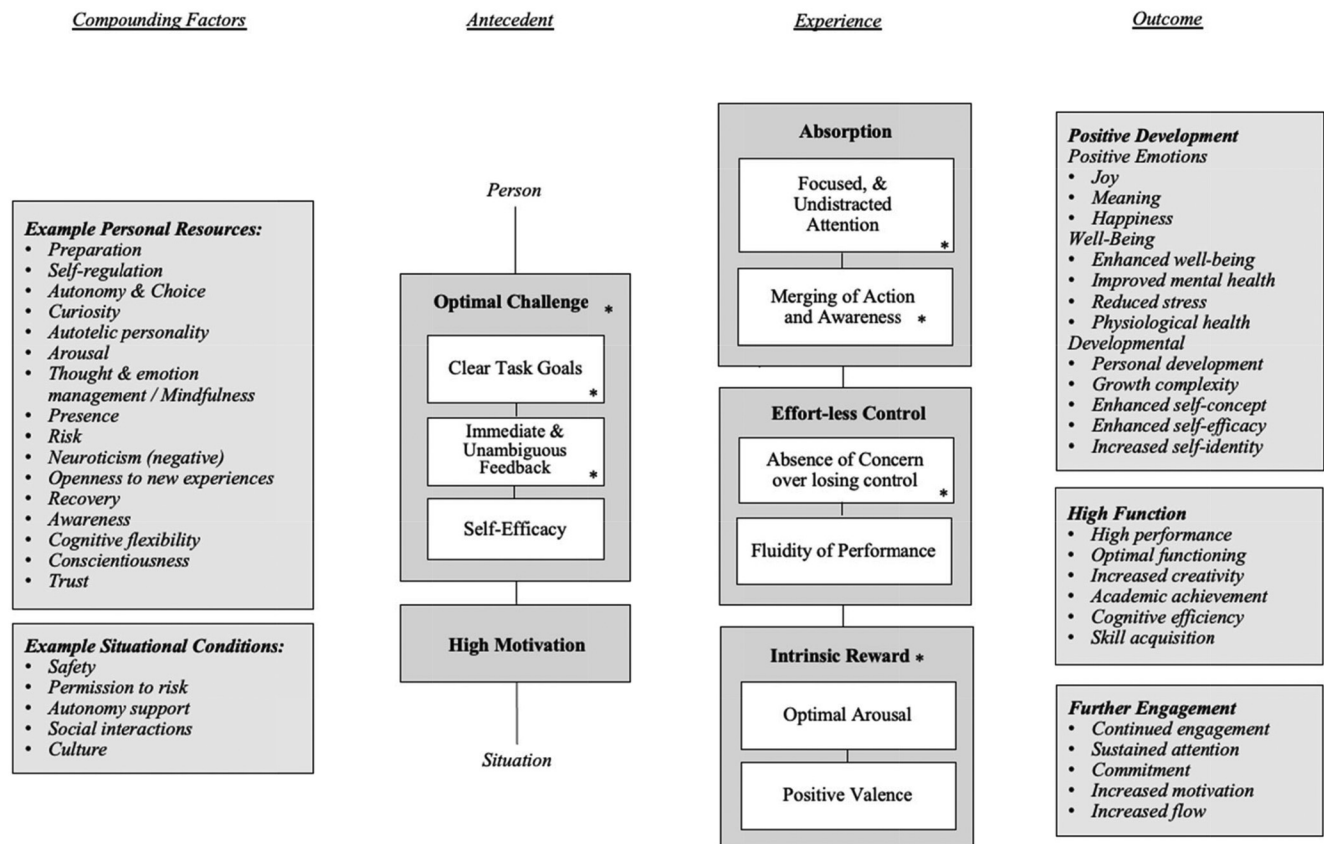
Flow Antecedent Dimension 1: Optimal Challenge

Sources most commonly identified Csikszentmihalyi's (1990) "challenge-skill balance" dimension as the primary (and frequently the only) antecedent (described: $k = 148$ /measured: $k = 52$) to flow. This dimension of the original model was frequently operationalized as an antecedent to flow in both correlational (for example, Bian et al., 2016; de Sampaio Barros et al., 2018; Flett, 2015) and experimental (for example, Tozman et al., 2017; Youssef, 2013) studies. Originally, Csikszentmihalyi (1975) contended that flow depends on establishing a balance between "action capabilities" [skills] and "action opportunities" [challenges] (p. 49), with the premise being that flow occurs when one is engaged in a challenge that stretches one's capability (Nakamura & Csikszentmihalyi, 2014). This "balance" or "stretching" between the *perceived* level of skill and challenge is proposed to be neither overmatching nor underutilizing (that is, matching at a 50:50 ratio above average levels; see Moneta, 2012; Nakamura &

Csikszentmihalyi, 2014). If the level of challenge exceeds the level of skill, then feelings of arousal and concern dominate one's experience. If the level of challenge is too low, however, then the individual's capability is not challenged and underarousal occurs (see Csikszentmihalyi's octant model of flow; Engeser, 2012a). Across studies, this optimal level of challenge was measured and described through a number of related terms ($n = 29$) such as "anxiety," "arousal," and "skills-demand fit," accounting for 31.29% of all utilized terms.

Psychophysiological evidence within the reviewed sources supported the notion that flow occurs when there is an optimal level of challenge and corresponding physiological arousal. Specifically, by examining ANS responses during flow (with participants engaged in tasks that challenged their ability but were not overly demanding), researchers reported an optimal (that is, moderate) level of activation in sympathetic activity (for example, Peifer et al., 2014; Tian et al., 2017), hypothalamic pituitary adrenal (HPA) axis levels (Bian et al., 2016; Peifer et al., 2014; Tozman & Peifer, 2016), high-frequency HRV (HF-HRV) and low-frequency HRV (LF-HRV), cortisol levels (Peifer et al., 2015), and respiratory depth (that is, larger breaths indicative of increased parasympathetic activity; Harmat et al., 2015). In other words, flow appeared to occur during increased parasympathetic modulation of sympathetic activity (Tian et al., 2017) and in an inverted U-shaped arousal (Peifer et al., 2014). This optimal (moderate) level reflects an inverted U-shaped relationship between challenge and flow (e.g., Tozman et al., 2017), suggesting that an individual first needs to be challenged or aroused, and that flow then occurs at an optimal level, around the peak of the inverted-U (when challenged but not overly so).

The literature examined in this review indicated that it may be most appropriate to term this antecedent dimension *optimal challenge*: a perceived capability to meet the challenging demands of the situation. The term optimal challenge subsumes constructs labeled by others as "challenge-skill balance," "level of challenge," "skills-demand fit," "skill level," "optimally challenged," and "skill integration," for example. The reasons for this relabeling are as follows. First, researchers have criticized the use of (original) terms such as "balance" or "50:50 ratio" because flow has been observed during instances of a slight imbalance between these concepts (for example, Engeser & Rheinberg, 2008; Lovoll & Vitterso, 2014; Vaughn, 2016). Similarly, anticipating flow in response to "above average" levels of these concepts may be conceptually ambiguous because what is considered "average" (skill or challenge) may differ between and within individuals and contexts (Keller & Landhausser, 2012; Yoshida et al., 2013; Mahnke et al., 2012)—the term optimal challenge instead suggests direction toward a positive high degree of challenge without limiting this degree to an absolute balance or discounting intra- or interdifferences. Second, sources within this review often measured the level of "challenge" and "skill" separately, and researchers subsequently made inferences about the degree of balance between those concepts. This is potentially problematic insofar as the terms challenge and skill may be partly conflated. Assessing the level of challenge may already acknowledge, to a degree, one's level of skill (for example, the level of challenge one faces is dependent, in part, upon the resources, or skills, one possesses). This separate assessment approach may result, therefore, in skill being measured twice, and may contribute to measurement inaccuracy (Keller &

Figure 2*Constructs, Compounding Factors, and Outcomes of Flow*

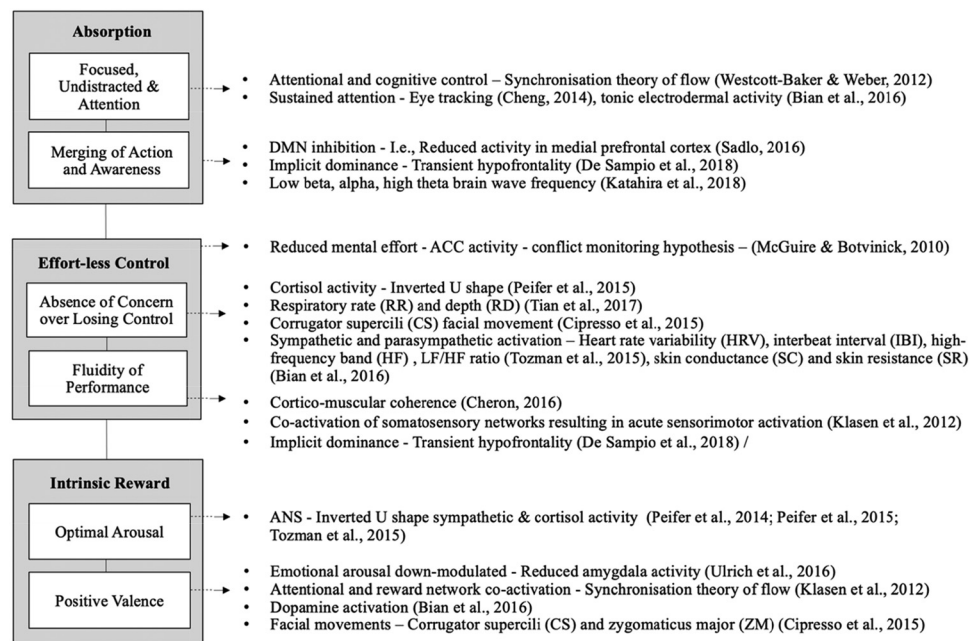
Note. ‘Optimal Challenge’ represents a more unidimensional construct than ‘challenge/skill balance’, which often measures challenge/difficulty and skill separately. ‘Total concentration on the task at hand’ has been updated to ‘focused and undistracted attention’. ‘Autotelic experience’ has been updated to ‘intrinsic reward’. ‘Sense of control’ has been more aptly defined by Csikszentmihalyi as an absence of concern or anxiety over losing control – summarized as ‘absence of doubt’. ‘Time transformation’ and ‘loss of reflective self-consciousness’ have been considered as variable experiences within ‘absorption’. * Csikszentmihalyi’s nine dimensions.

Landhasser, 2012; Mahnke et al., 2012). In addition to issues of conflation, the independent measurement of challenge has, at times, also proved difficult, such as those instances when researchers have used performance-related markers of success (e.g., Klasen et al., 2012).

The evidence in this review indicated that this antecedent dimension may incorporate three subthemes that are commonly referred to in flow research, and that were often used to help operationalize this dimension—namely *clear task goals*, *immediate and unambiguous feedback*, and *self-efficacy*. The construct clear task goals subsumes the example terms (utilized by review sources) including “specified objectives,” “clarity of purpose,” “confidence”; “immediate and unambiguous feedback” subsumes the example constructs “coherent, noncontradictory demands” and “direct feedback”; and “self-efficacy” subsumes the example constructs “self-confidence” and “self-trust.” See Figure 2 and S1, tab Thematic Grouping of Terms for further details. The premise behind goal- and feedback-related concepts being included in this theme appears to be that if goal (that is, clear goals) and diagnostic information are not clear and immediate (that is, immediate and unambiguous feedback), then the experience may be confusing or

overdemanding (and no longer optimally challenging; Cseh, 2016; Drengner et al., 2018; Guertin, 2016; Keller & Landhausser, 2012; Moneta, 2017; Simles et al., 2018; Valenzuela et al., 2018). When examining goal type, sources reported that goal congruency (for example, Keller & Landhausser, 2012), personal growth and exploration goals (for example, “seeing how well I can do”; Swann et al., 2016), “open” (that is, nonoutcome orientated) as opposed to “fixed” (that is, outcome orientated) goals (Swann et al., 2017), and task-orientated goals (Csikszentmihalyi, 2014c) result in increased flow. Finally, with respect to the self-efficacy subtheme, it has been recognized for some time that “perceived ability,” and more recently, self-efficacy (for example, Cheng, 2014; Chen & Sun, 2016; Salanova et al., 2014), acts as a facilitator of flow. Waples and Knight (2017), for example, described that self-efficacy and flow may be related in a way that mimics an “upward spiral” effect. In terms of its positioning within this antecedent theme, self-efficacy is considered a personal (“capacity” or “control”) resource that buffers against stress and has resilient properties (Bandura, 1986), thereby supporting individuals’ capabilities to meet challenging situations more effectively (Keeler et al., 2015).

Figure 3
Physiological and Neuroscientific Measures and Markers to Flow



Note. Certain markers may better reflect the experience (as a whole) rather than specific dimensions or subdimensions.

Flow Antecedent Dimension 2: High Motivation

Review sources also brought to light a second broad antecedent dimension—capturing concepts relating to high motivational forces—which was especially valuable when examining the occurrence of flow in nonachievement settings (e.g., Engeser & Schiepe-Tiska, 2012). The construct “high motivation” subsumes the example terms (utilized by review sources) “importance,” “intrinsic motivation,” “and “interest” (e.g., Bricteux et al., 2017; Csikszentmihalyi, 2014c; Rheinberg & Engeser, 2018). See Figure 2 and S1, tab Thematic Grouping of Terms for further details. Authors of reviewed sources posited that flow was more likely to occur when individuals reported or displayed a high level of motivation for the activity (e.g., Boffi et al., 2016; Geyser et al., 2015; Llorens et al., 2013; Schüler & Brandstätter, 2013). For example, de Sampaio et al. (2018) concluded that “flow feelings are highly connected to conditions that promote individuals’ choice and autonomy” (p. 810). Baumann (2012) also proposed that the achievement flow motive operationalizes an autotelic personality toward flow; and, Engeser and Schiepe-Tiska (2012) proposed that flow emerges from the interaction of one’s motives and the motive-specific incentives available in the situation. Accordingly, instances where flow was described as a function of “importance,” “motivation,” “reasons for participation,” “interest,” “curiosity,” and “autonomy” were all captured within this dimension. Together, these terms accounted for 4.59% of all terms and 34% of the total measured antecedents.

The occurrence of flow has long been associated with intrinsic motivation (Bakker et al., 2017) due in part to the intrinsically rewarding nature of the experience and Csikszentmihalyi’s

discussion of “autotelic experience” (see Csikszentmihalyi, 2014b). Accordingly, sources posited that task interest (Bricteux et al., 2017; Cheng, 2014; Culbertson et al., 2015; Llorens & Salanova, 2017; Shernoff et al., 2014), “outcome importance” (Abuhamdeh & Engeser, 2012), “task importance,” “goal congruency” (e.g., Keller, 2016), “motivational regulatory orientations” (e.g., Keller & Landhausser, 2012), and “subjective value” predict the likelihood and intensity of flow (also see Csikszentmihalyi, 2014b, 2014c; Keller & Landhausser, 2012). For example, Auld’s (2014), review concluded that *interest* in content and subject was an integral antecedent for finding flow in learning, and Keller and Landhausser (2012) propose that the *subjective value* predicts and explains the variations between micro and macro (i.e., deep and mild) experiences of flow—which to date, have only been explained through the escalating balance between challenge and skill (i.e., the higher the challenge and skill level the higher the flow intensity). However, reviewed sources also highlighted that extrinsic motivation concepts may play a pivotal role in the occurrence of flow (e.g., Schüler & Brandstätter, 2013; MacNeill & Cavanagh, 2013; Valenzuela et al., 2018; Vaughn, 2016; Vuorre & Metcalfe, 2016). For example, Bassi and delle Fave (2012) reported work-related flow to be associated with controlled regulation, and Vaughn (2016) reported that NFL players reported high levels of extrinsic (as well as intrinsic) motives in their most memorable flow experiences. In fact, extrinsic factors leading toward flow are so common in work settings that in some flow literature this is known as the “paradox of work” (e.g., Bassi & delle Fave, 2012).

In summary, although research has consistently linked flow experiences with a high degree of motivation (e.g., Boffi et al.,

2016; Geyser et al., 2015; Llorens et al., 2013), further research is needed in order to better understand the specific types of motivation that are most conducive to flow experiences and how they relate, or not, to an “optimal challenge.”

The themes we described above were those that, on the balance of evidence, appeared to be considered (and act) as antecedents, or important preconditions, for flow experiences. These themes mirror, in some respects, elements of the most widely-used (nine-dimensional) flow model (i.e., challenge-skill balance, clear goals, unambiguous feedback), but in this instance it is important to note that they were positioned as precursors to (rather than components of) flow experiences. We did, however, also observe some consistency in terms of the characteristics and components of flow experiences, and in the sections that follow we overview the three themes that we observed in this respect.

The Flow Experience Dimension 1: Absorption

The evidence we reviewed demonstrated that an absorption in the task—characterized by focused and undistracted attention (i.e., total concentration; see Moran, 2012; Ulrich et al., 2016a) and a merging of action and awareness—is a key experiential component of the flow experience (Csikszentmihalyi, 2014a). For example, Nakamura and Csikszentmihalyi (2014) explained that flow involves an intense and focused concentration on what one is doing in the present moment, resulting in a merging of action and awareness. Scholars have argued that during this absorption, attentional resources that ordinarily facilitate reflective self-consciousness (i.e., the hyper egoic self or Mead’s “Me”) and temporal experience are instead allocated to present attentional demands, resulting in a “loss of reflective self-consciousness” and “time distortion” (e.g., Nakamura & Roberts, 2016). However, scholars have also pointed out inconsistencies with the existing dimensions “loss of reflective self-consciousness” and “time transformation” (see Boyd et al., 2018; Swann et al., 2012; Vaughn, 2016). For example, an individual in flow may also be acutely aware of time if the activity requires, such as running down the clock in a basketball match. Correspondingly, 72 terms accounting for 34.44% of the total described and measured terms were synthesized into this theme. Commonly-used terms in the flow literature—and that are recognizable from the nine-dimensional model—that were coded in this theme included “total concentration,” “loss of self-consciousness,” “merging of action and awareness,” and (in some instances) “time transformation.” Parenthetically, it is also worth noting that “absorption/immersion” is often referred to using different labels across subdomains, such as “telepresence” in human computer interface or gaming (e.g., Guertin, 2016; Klasen, 2012; Lazoc & Luiza, 2012; Miles, 2012). Additionally, we found sources frequently associated this dimension (absorption) with a deeper sense of “connection” (e.g., Csikszentmihalyi, 2014a; Garcés-Bacsal, 2016) to equipment, people, or other components relevant to the task (e.g., “being at one with the activity,” “self-transcendence,” “connection,” “sense of unity,” “shared identity”). Reports are preliminary, however, and further research is required to identify reliable characteristics of such a phenomenon.

Many researchers highlighted that absorption is critical to the flow experience (e.g., Drengner et al., 2018; Fullagar & Kelloway, 2012; Harris et al., 2017), and this notion is reflected in multiple flow instruments (e.g., FSS, Rheinberg et al., 2002; WOLF,

Bakker, 2008; EGameFlow, Fu, et al., 2009). In addition, authors also noted that failing to account for this dimension (i.e., absorption) in measurement efforts may produce misleading conclusions (Schiefele, 2013), because it is this concept that differentiates flow, at least in part, from other states of concentrated action (e.g., Csikszentmihalyi, 2014a; Sheldon et al., 2015). This differentiation is perhaps best illustrated by those sources in which brain wave frequencies were examined (through EEG measures). Specifically, whereas high attentional states of control are associated with high beta frequencies, flow experiences have been shown to display varying theta, moderate alpha (Katahira et al., 2018), and low beta (Pluta et al., 2018) activity, which is typically more closely representative of trance-like immersive states.

The sources in this review widely supported the notion that absorption is a defining characteristic of flow; however, researchers differed, at times, in how they positioned this theme relative to (or within) Csikszentmihalyi’s nine dimensions. For example, some authors grouped Csikszentmihalyi’s dimensions “merging of action of awareness,” “loss of (reflective) self-consciousness,” and “time transformation” as related but independent subcomponents of “immersion” (e.g., Fang et al., 2013; Sillaots & Jesmin, 2016; Simlesa et al., 2018; Yoshida et al., 2014). Alternatively, others identified absorption through measures of time transformation-only (e.g., Grigorieva, 2015), or categorized “loss of reflective self-consciousness” and “time transformation” as outcomes of the flow experience (e.g., Herrando et al., 2018). Overall, the sources we reviewed revealed a trend toward measuring absorption as a single construct, accounting for “established” flow concepts “merging of action of awareness,” “loss of reflective self-consciousness,” and “transformation of time” (for example, Bakker et al., 2017; Fullagar & Kelloway, 2012; Llorens & Salanova, 2017; and, although not included in this review, see also Engeser & Rheinberg’s, 2008; conceptualization of flow as “absorption” and “fluency of performance”). The premise behind this simplification appears to be that when attention is wholly invested in and directed toward a task, action and awareness merge in the absence of spare attention, and self-related thoughts and monitoring of experiential factors (such as reflective self-consciousness) fade from awareness (Csikszentmihalyi, 2014b; Nakamura & Csikszentmihalyi, 2002; Pelet et al., 2017). Interestingly, part of the support for this absorption theme came from a branch of neuroscientific research, in which it was suggested that flow occurrence results in a depleting “onion-peeling” effect of higher cognitive processes (e.g., Stoll & Pithan, 2016; Ulrich et al., 2016b). During flow, it is argued by some that the highest layers of consciousness—those that include higher cognitive functions (i.e., self-reflective thinking and time monitoring)—are most susceptible to change when alterations to mental states occur and resources are reallocated (Stoll & Pithan, 2016). Accordingly, the more intense the flow experience, the greater number of higher cognitive functions that are down-regulated. Evidence indicates that flow is associated with (a) reduced activity in the medial prefrontal cortex (a neuroanatomical area for reflexive conscious thinking; e.g., Ulrich et al., 2014); (b) inhibited activity in a group of brain regions called the default mode network (also known as self-awareness network; Sadlo, 2016; Ulrich et al., 2014; Ulrich et al., 2016b); and (c) a down-regulation of the explicit cognitive system (i.e., volitional and conscious skills) as outlined in transient hypofrontality theory (see Dietrich, 2004; Nakamura & Roberts, 2016). This depleting

“onion-peeling” effect may shed light on the neuroscientific mechanisms that explain the reliability inconsistencies of certain measured constructs from Csikszentmihalyi’s nine dimensions (e.g., “loss of self-consciousness” and “time transformation”).

The Flow Experience Dimension 2: Effort-Less Control

In addition to the notion of absorption, there was evidence that flow experiences are also characterized by a degree of “effortless control”—a high sense of control in which the task feels less effortful than is typical for that person, characterized by fluidity of performance and an absence of concern over losing control. For example, In Beard’s (2015) interview, Csikszentmihalyi describes the effortless behavior in flow as “the kind you do not have to think about it, you’re just doing it” (p. 358); or, more recently, Csikszentmihalyi et al. (2018) described flow as “an almost automatic, effortless, yet highly focused state of consciousness” (p. 216). To illustrate, 66 terms accounting for 12.81% of the total described and measured terms were synthesized into this dimension. Commonly-used terms in the flow literature that were coded in this theme included “sense of control,” “fluency,” and “effortless involvement.”

A sense of control, irrespective of how it was defined, was the most commonly used descriptor of flow in this review. This sense of control—or, as Csikszentmihalyi (2014a) more recently described, “a lack of anxiety about losing control” (p. 231) and not a strained conscious control that may otherwise occur from an increase in anxiety over losing control—is considered integral to the unusual fluidity of action and felt effortlessness (e.g., Bian et al., 2016; Järvillehto, 2016) that identifies flow from other high-performance acts (i.e., clutch states; Swann et al., 2017). For example, in a meta-analytic review within computer-mediated learning, Auld (2014) stated that “flow enables the individual to enjoy the activity in a seemingly effortless manner... feeling in control of one’s actions during a task, despite a seeming automaticity and effortlessness to one’s behaviors; hence participants have described this feeling as if they were “flowing” on a current” (p. 6). Equally, in neuroscientific or psychophysiological studies, flow is frequently described as a subjective state of effortlessness (e.g., Bian et al., 2016) in which attention is effortless (e.g., de Manzano et al., 2013). In the literature on neurophysiology, subjective effort-less control has been aligned with fast and efficient physiological processes that are internally coherent and harmonious. Specifically, when experiencing effortless control, there appears cortico-muscular coherence (e.g., Cheron, 2016), coactivation of brain regions (e.g., Weber et al., 2016), balanced ANS activity (e.g., Tozman et al., 2017), and synchronized changes in neural activity (i.e., inferior frontal gyrus, left putamen, and amygdala) that permit the psychophysiological processes of a deeper sense of cognitive control, increased coding of outcome probability (putamen), and decreased negative arousal (e.g., Ulrich et al., 2014; Ulrich et al., 2016b).

Flow has also been associated with a reduction in conflict-monitoring cognitive processes as evident by reduced activity in the anterior cingulate cortex—an area of the brain designed to assign effortful resources when conflict is assumed (see conflict-monitoring theory: McGuire & Botvinick, 2010). Additionally, an inhibition of the posterior cingulate cortex (a core region of the default mode network; Huskey et al., 2018) and explicit cognitive system

(i.e., transient hypofrontality; see Nakamura & Roberts, 2016), which results in a reduction of concern about self-ability and allows for more efficient and fluid processing and movement, is highlighted to be apparent during flow (Bian et al., 2016). In EMG and EEG studies, high levels of cortico-muscular coherence suggest that flow is linked with a synchronization of neural activity between the brain and body, which is responsible for the high level of control and, more precisely, the smoothness perception or fluidity of action that is described in flow (see Cheron, 2016). In further examination of the high degree of felt control in flow, flow is linked with synchronized activity in somatosensory networks and acute sensorimotor activation, which facilitates heightened feedback and a subsequent increased sense of control (Klasen et al., 2012).

In regard to flow feeling less effortful than typical, the synchronization theory of flow outlines how a highly efficient “switching on” of networks associated with goal-directed activity, and a corresponding “switching off” (due to limited capacity) of areas related to the self and perception of effort reduces the effort felt, even during high physiological effort (see Westcott-Baker & Weber, 2012; Huskey et al., 2018). In adjacent research, authors have highlighted an inhibition in debilitating emotional responses that may require effort to overcome. For example, some scholars have highlighted that the heightened concentration during flow is posited to temporarily exclude all irrelevant thoughts and feelings from consciousness (Csikszentmihalyi, 2014a; Guertin, 2016; Swann et al., 2017). Concurrently, neuroscientific and psychophysiological researchers indicate that a defining feature of flow is the reduced emotional activity that is apparent in (a) the down-modulation of emotional arousal mediated by the amygdala (Ulrich et al., 2016b); (b) a down-regulation of medial prefrontal cortex, which plays an important role in emotional biasing during decision-making (Nah et al., 2017); and (c) reduced activity in the dorsomedial prefrontal cortex, which is responsible for activity surrounding self-related emotions (Nah et al., 2017). Further research is required to understand the psychological ramifications of these neuroscientific findings, though this preliminary research helps to shed light on how flow can feel effort-less even during objectively effortful acts.

In summary, this effort-less control, or reduced felt effort deriving from a reduction of mental conflict and increased fluidity of movement, is seen as a principal characteristic of flow because traditional models of focused engagement otherwise require high degrees of felt effort (i.e., cognitive/mental effort and arousal) to override external distractors (Gaggioli et al., 2013; Harris et al., 2017; Peifer et al., 2014; Romero & Calvillo-Gamez, 2014; Tozman et al., 2015). Järvillehto (2016) argues that it is precisely this reduced effort and inhibition of the explicit system that allows intuitive rather than self-reflective action, and posits that it is this intuitiveness of the behavior that allows such effortless control to be felt.

The Flow Experience Dimension 3: Intrinsic Reward

This review found a diverse range of constructs representing flow as an intrinsically rewarding (i.e., “autotelic”) experience that is characterized in a psychophysiological way by positive valence and optimal levels of arousal. For example, the majority of sources pointed toward a sense of enjoyment or intrinsic reward during

flow as a fundamental characteristic of flow (e.g., Nakamura & Csikszentmihalyi, 2014; Valenzuela et al., 2018; Yoshida et al., 2013). Correspondingly, 28 terms accounting for 9.77% of the total described and measured constructs were synthesized into this construct, including the common terms employed in flow literature: “autotelic” (e.g., Csikszentmihalyi, 2014a), “joy” (e.g., Chilton, 2013), “enjoyment” (Llorens et al., 2013), “positive affect” (e.g., Omori et al., 2014), “intrinsic reward” (e.g., Romero & Calvillo-Gamez, 2014), “positive valence” (e.g., Peifer et al., 2014), “pleasure” (e.g., Lazoc & Luiza, 2012), and “satisfaction” (Wolfiel & Czerw, 2017).

An autotelic experience was the seventh most described ($k = 141$) and ninth most measured ($k = 70$) term of flow, likely due to its inclusion in the nine-dimensional model. It is unclear, however, whether Csikszentmihalyi meant for an autotelic experience to represent a holistic description of the entire flow experience (flow was originally labeled autotelic experience; e.g., Abuhamdeh & Engeser, 2012), a part of the experience (e.g., Araújo & Hein, 2016; Kaur et al., 2016), a consequence of flow (e.g., Perttula et al., 2017; Schiefele, 2013; Swann et al., 2018), or a reason for participation (e.g., Corbato, 2018; Monson, 2013). Accordingly, authors utilized this autotelic component differently. The terms intrinsic reward (e.g., Nakamura & Csikszentmihalyi, 2014) and enjoyment (e.g., Miles, 2012) were instead used more consistently, with most scholars positioning them as a component of the flow experience. From a neurobiological perspective, there is a strong link to attention and reward such that the sustained attention in flow requires the experience to be somewhat rewarding, as is evident in the activation of midbrain reward structures (Nah et al., 2017) and increased dopamine production during flow (Bian et al., 2016). Dopamine is considered to be an essential neurochemical in the brain’s reward system, making the flow experience feel highly rewarding. In fact, the correlation between flow and dopamine is considered so strong that flow proneness has been related to a polymorphism of the dopamine D2 receptor coding gene (DRD2 C957T rs6277) which affects striatal D2 receptor availability and thus an individual’s ability for flow proneness (Gyurkovics et al., 2016). For example, Huskey et al. (2018) posited in the synchronization theory of flow (i.e., synchronization process between structures within cognitive control and reward networks) that intrinsic reward is biologically inevitable during flow, and that fMRI studies show that optimal levels of challenge corresponded to the highest levels of intrinsic reward (when compared with problems that are too easy or difficult).

Lastly, of note, is the importance of differentiating the *real-time* intrinsic reward felt during the experience and emotions (see also Abuhamdeh, 2020) that accompany one’s reflective assessment of the event—the latter may be a better reflection of the participant’s immediate *response* to the experience (e.g., enjoyment, joy, surprise, happiness; Llorens et al., 2013); therefore, the term intrinsic reward rather than enjoyment was deemed a more accurate reflection of this dimension. Concurrently, experimental studies have shown that flow experiences are characterized by optimal levels of arousal, and high values for positive valence that then lead to positive affective states (see Cheng, 2014; Peifer et al., 2014; Ulrich et al., 2016b). Equally, emotion theorists (e.g., Barrett et al., 2007; Moors, 2009) often describe emotion as a composite of two dimensions, valence and arousal, that combine to produce an “affective quality” or emotion (Moors, 2009) such as enjoyment (subsumed

in this dimension). Given this dimension encapsulates the emotional term “enjoyment,” we thought it best to position arousal and valence indicators under the dimension “intrinsic reward.” Further research is required, however, to understand whether arousal markers best represent this dimension, another dimension, or all three dimensions (i.e., the experience as a whole).

In terms of understanding those constructs that arise as a result of flow, rather than those that describe the flow experience itself, this review revealed three common outcome themes (which we present below).

Flow Outcome Theme 1: Positive Development

Positive development accounted for 45.6% of total reported outcomes of flow and was the most commonly studied consequence of the construct. Emotional, well-being, health, and developmental outcomes were all synthesized into this theme, with specific experiences such as joy, meaning, satisfaction, happiness, personal development, growth complexity, enhanced self-concept, increased self-identity, physiological health, mental health, and reduced stress all linked to flow. In the event of further causal research shedding light on flow outcomes, researchers may find it more appropriate to position developmental and well-being (i.e., health and emotion) outcomes as distinguishable subcategories, or as separate outcome themes.

One rationale for an association between flow and positive development was offered by Nakamura and Roberts (2016); these authors argued that self-reflective attention, which is absent during flow, is chiefly responsible for many self-defeating processes such as anxiety, stress, negative affect, negative health, and the causation of depressive tendencies. Even at a basic level, self-focused attention can impair the quality of one’s experience and limit attentional capacity causing moment-to-moment attentional conflict, effort, and subsequent stress. Because flow facilitates an absorption that inhibits this self-reflective activity, attentional conflict and stress is reduced, which subsequently allows room for positive outcomes (e.g., Nakamura & Roberts, 2016). For example, Fullagar et al. (2013) reported that flow—as operationalized through challenge-skill balance and measured through an adaptation of the FSS-2 scale—and task-specific anxiety were antithetical experiences such that when flow was highest, performance anxiety was lowest (and vice versa). These intrinsically rewarding experiences of effort-less control result in a more ordered self, stronger self-efficacy (e.g., Waples & Knight, 2017), personal satisfaction (e.g., Miles, 2012) and ultimately improved well-being and health (e.g., Magyarodi & Oláh, 2015). For example, Riva et al.’s (2016) review indicated that flow in everyday life was connected with individuals’ well-being, reduced symptomatology, and positive and meaningful changes both inside and outside the therapeutic and rehabilitation processes. Developmentally, Freire et al. (2016) posited that these flow experiences can also be referred to as “optimal leisure,” which facilitate several developmental functions and positive personal and social skills and achievements. Overall, flow has been viewed as an effective and rewarding coping mechanism for anxiety- and stress-reduction when facing daily-life challenges (e.g., Llorens & Salanova, 2017; Sadlo, 2016), associated with moderate cortisol levels (i.e., not associated with high levels of cortisol that might ordinarily be associated

with distress) during challenging task engagements (e.g., Peifer et al., 2014), and for deriving potential health benefits.

Flow Outcome Theme 2: High Functioning

A theme relating to “high functioning” (28.44%) brought together terms such as “high performance,” “optimal functioning,” “increased creativity,” “academic achievement,” “optimal learning,” “increased productivity,” “cognitive efficiency,” “skill acquisition,” and “better judgments.” For example, Andersen (2016) associated flow, as measured through ESM in combination with observations and interviews, with better STEM learning. Further, Flett (2015) found that challenge-skill balance was correlated with positive functioning and led to higher performance ratings while playing tennis, and Wang and Hsu (2013) reported that flow improved learning performance and learning satisfaction. Although the majority of sources reported perceived functioning or performance as outcomes of flow (e.g., Swann et al., 2017), some authors considered flow to be synonymous with objective performance success (e.g., Klasen et al., 2012). Given that peak performance (e.g., winning a sporting event) is dependent in many cases on external influences outside of an individual’s control (Privette, 1983), however, performance success may be best considered as a possible (rather than expected) outcome of the construct.

Flow Outcome Theme 3: Further Engagement

Finally, a theme emerged reflecting “further engagement” (19.94%), which accounted for (a) continued engagement, adherence, and ongoing attentional recourses allocated to the relevant task in real time (e.g., Nakamura & Roberts, 2016), and (b) a temporally delayed reengagement (e.g., Csikszentmihalyi, 2014a). Terms such as “continued engagement,” “sustained attention,” “commitment,” “exercise adherence,” “optimal engagement,” “further flow,” and “increased motivation” were included in this theme. Sources considered flow to be a highly rewarding experience that led individuals to be naturally drawn toward reengagement even at great risk (Csikszentmihalyi, 2014b). For example, Nakamura and Csikszentmihalyi (2009) described emergent motivation, meaning that once flow is achieved, the emergence of new intrinsic motives facilitates further engagement. In an application of this principle, Rodriguez-Ardura and Meseguer-Artola (2017) reported that flow, as measured through a self-report questionnaire (examining challenge level, control, focused attention, presence, and positive affect), enhanced e-learners’ academic performance and contributed to students’ effective continuance in e-learning. Further, Harris et al. (2017) reported that flow resulted in continued attentional effort and allocation of attentional resources—which has both biological and sociological ramifications.

Discussion

Given the variation in conceptualizations of flow and the divergence in methods used to operationalize the construct, flow research is considered by some to have reached “crisis point” (e.g., Swann et al., 2018). Guided by scoping review recommendations, the broad purpose of this study was to bring together—and chart, in one place—the varied conceptualizations of flow that have appeared in the literature since 2012. We also sought, if

possible, to identify commonalities in flow conceptualization that may provide a more coherent platform for future research in the area. Our review of over 200 sources demonstrated that the study of flow remains a popular research topic—flow research has been undertaken (since 2012) by researchers from 37 countries (predominately in Europe and the United States) and has appeared within 48 different scientific disciplines, including cognitive neuroscience, art therapy, sociology, clinical psychology, electrophysiology, artificial intelligence, human computer interface, and neuropsychophysiology. Unsurprisingly, flow research has been most frequently published in the broad field of psychology (accounting for 73% of studies in this review), especially in positive psychology (18%), sport and exercise psychology (10%), occupational psychology (7%), educational psychology (6%), and personality psychology (4%) fields. Importantly, although the review in part highlighted the disparate nature of flow research (e.g., 150 different constructs, terms, or concepts were used to describe the flow experience, and 84 different terms were positioned as flow antecedents), some commonalities also emerged with respect to flow antecedents, characteristics, and (to a lesser extent) outcomes. In the material that follows, we consider key implications of these findings.

The Conceptualization and Measurement of Flow

This review identified two commonly reported or studied antecedent themes (optimal challenge and high motivation) and three experiential themes (absorption, effort-less control, and intrinsic reward), with the experiential constructs associated with psychological and physiological markers (see Figure 3 and the Results section for a full analysis). With respect to the experience of flow itself, review sources indicated that flow is characterized by focused and undistracted attention, a merging of action and awareness, fluidity of performance, an absence of concern over losing control, positive valence, and optimal levels of arousal (see Table 1). It appears, therefore, that (according to the sources reviewed in this study at least) an appropriate way to define flow may be as follows: *flow is an intrinsically rewarding state of absorption in a task in which control feels effort-less*. In addition to evidence for common antecedents and experiential concepts, three broad outcomes of flow were highlighted or studied—namely, positive development, high functioning, and further engagement.

This review is important not least because it addresses recent calls to reduce (unnecessary) theoretical complexity, eliminate redundancy, and strive for conceptual parsimony in psychological research (e.g., Hagger, 2014; Le et al., 2010). Specifically, compared with the somewhat inconsistent application (and in some cases, conceptualization) of the nine-dimensional model, the two antecedent and three “flow experience” themes that emerged in this review may provide greater clarity for those seeking to understand, measure, and facilitate flow (and associated mechanisms or processes). In addition, and with those calls in mind, researchers have also been encouraged to consider the nature of flow relative to other related constructs, such as “peak experience,” “ecstasy,” “work engagement,” “immersion,” “peak performance,” “presence,” “intrinsic motivation,” “mindfulness,” “cognitive absorption,” or “deep play.” The results of this review indicate that there is clearly a continued research interest in, and attraction to, the

Table 1
Flow Dimensions and Descriptions

<i>Proposed definition of Flow</i>	Flow is an intrinsically rewarding state of absorption in a task in which control feels effort-less
<i>Antecedents to flow</i>	
Optimal challenge	A perceived capability to meet the challenging demands of the situation
High motivation	A high motivational force
<i>Experiential dimensions to flow</i>	
Absorption	A state of absorption in the task characterized by focused, undistracted attention, and a merging of action and awareness
Effort-less control	A high sense of control in which the task feels less effortful than is typical for that person, characterized by fluidity of performance and an absence of concern over losing control
Intrinsic reward	An intrinsically rewarding experience characterized by positive valence and optimal levels of arousal

Note. Bold terms indicate categories.

flow construct—an interest and attraction that cuts across research methodologies and disciplines. This continued focus on flow may be in part due to the nature (i.e., the quality) of the experience itself. That is, although other related constructs may play a role in (and share elements of) the experience of flow, it appears that the combination of absorption, effort-less control, and intrinsic reward makes flow appealing from both a practical and a scientific perspective. It is interesting to note, for example, that individuals rarely focus on singular elements or components when describing flow. Most often, and as was observed in many of the sources reviewed in this study, it is the combination of (i.e., absorption, effort-less control, and intrinsic reward) elements that characterize individuals' descriptions of a latent flow experience. In addition to providing greater clarity on the psychological dimensions of flow, this review also helped identify the “nonpsychological” markers associated with flow experiences (see Figure 3). By integrating recent neuroscientific and physiological studies with those focused on people's experiences of flow, these findings provide a foundation for future research that intersects neurological, psychological, and physiological perspectives on the construct.

With an emphasis on the measurement of flow experiences (and antecedents), we observed substantial variation and a lack of consistency in the instruments used within the studies in this review. In addition to identifying a plethora of measurement instruments, it was apparent that the most commonly used instruments have attracted at least some criticism (see Moneta, 2012; Swann et al., 2018). It was also apparent that no existing instrument (a) adequately assesses all of the dimensions of flow outlined in this review; (b) is able to identify when a person is “in” or “out” of flow (Engeser, 2012b); or (c) offers a clearly superior way of measuring the intensity of the flow experience (Jackman et al., 2017; Moneta, 2012). Regarding the issue of identifying the presence or absence of flow, researchers are encouraged to move beyond identifying flow when summed mean dimensional scores are above a midpoint (often labeled, *neither agree or disagree*) as low dimensional scores may better reflect nonflow states (Jackman et al., 2017). Further, accounting for multiple occurrences of flow within a single activity or defined time period may be more accurate. One potential solution in this respect may be provided by Ellis et al.'s (2019) suggestion to integrate Hsiang et al.'s (2000) binary method of measurement with Csikszentmihalyi's original flow questionnaire methodology. Such a method could be administered postactivity (or through ESM), whereby participants are asked to respond “yes” or “no” (e.g., binary response instead of, or

as well as, a continuous response scale) to a description of flow and its dimensions. This method may provide insight into whether focal individuals consider themselves to be experiencing (or to have experienced) flow. Meanwhile, a more detailed assessment of flow intensity and frequency could be achieved through questions focused directly on soliciting participant ratings of these descriptors. Respondents might be asked, for example, “Can you relate to experiencing this description?” (yes/no), and if yes, to identify the strength or intensity of the experience, and whether it reflected one continual experience or was a case of entering and exiting the experience multiple times. If indicating the latter, participants may also be asked to indicate the proportion of the total time they were in flow within the activity and add commentary on flow restoration or other temporal factors. Such a solution could be compared against the common approach of measuring flow through a Likert scale instrument (continuous measure) to ascertain possible thresholds of flow entry.

Integration of Postreview Research

Flow research has continued to accumulate since the period during which this review was conducted. It is important, therefore, to highlight certain important contemporary works that offer insight into the fundamental conclusions of this review. Most notably, Peifer and Engeser (2021a) edited a recent update to the 2012 collection of articles on flow (Engeser, 2012a), and in support of our review findings concluded that (a) “a gold standard for the modeling and measurement of flow is not at close reach” (p. 61)—highlighting that measuring flow is in need of continuous improvements in line with theoretical advancements; and (b) there was support for synthesizing flow descriptors into three core experiential dimensions (absorption, perceived demand-skill balance, enjoyment). Peifer and Engeser's (2021b) work also synthesized a number of flow characteristics into a core dimension, labeled “absorption,” to represent the high level of control felt in flow. In addition, Peifer and Engeser (2021b) labeled a secondary core dimension as “perceived demand-skill balance.” An important differentiating approach between the findings of this review and Peifer and Engeser's (2021b) labeling of “perceived demand-skill balance” is that sources within this review highlighted (both psychologically and neuroscientifically) that it is the sense of “effortlessness” (i.e., a subjective sense of the act being less effortful or more fluid than usual) toward the sense of control that differentiates flow from other forms of high control (also see

Peifer & Tan, 2021). Additionally, we observed that review sources in our work recognized the optimal level of challenge dimension (i.e., perceived demand-skill balance) as an antecedent to flow (e.g., Barthelmäs & Keller, 2021)—one that should be separated, therefore, from the experience of a high effort-less sense of control. Another important differentiating approach between this review and Peifer and Engeser's (2021b) suggested core dimensions, is that although both the findings of this review and Peifer and Engeser's (2021b) work aimed to encapsulate the autotelic experience of flow, we labeled that dimension as "intrinsic reward," differing from Peifer and Engeser's (2021b) utilization of the term "enjoyment." In this review, sources highlighted that to acknowledge an experience as "enjoyable" (or not) involves reflective cognitive processes that are deemed open to bias from outcomes and contextual or social factors. To illustrate, one may experience an intrinsically rewarding hit of dopamine during an optimally aroused state, for example, yet still reflect on the experience as not enjoyable if they lost the sporting contest or received negative feedback from a colleague (see Abuhamdeh's, 2021; review on the relationship between flow and enjoyment). The term intrinsic reward, therefore, was deemed as a more appropriate depiction of a flow-like experience.

In terms of understanding flow intensity, frequency, and duration, there is also a long-standing debate about whether flow is a discrete (i.e., in-or-out) or continuous (i.e., mild-to-intense) state. Although most flow researchers measure flow with Likert-type scales (thereby assuming flow is a continuous state), Abuhamdeh (2020) suggested that, if flow is conceptualized as an "optimal" experience, it should be measured as a discrete state and not in terms of varying levels of intensity. That is, Abuhamdeh argued that lower intensities are unlikely to be optimal, and that flow can only be experienced once a high threshold is met. Peifer and Engeser (2021b), on the other hand, suggest that flow can be both a discrete *and* continuous state; an experience can be considered "flow" when certain markers above a threshold are met (i.e., discrete), then "the intensity of the experience becomes even more intense (i.e., continuous) the more the components are present and pronounced" (p. 425). Further clarity in this area is clearly required in order to develop more effective forms of measuring (and promoting) flow.

Postreview neuroscience and psychophysiology studies continue to support the flow components of absorption, effortless control, and intrinsic reward. Three publications, however, offer additional insight beyond the review findings. First, van der Linden et al. (2021) positioned that the higher dopaminergic activity that makes flow intrinsically rewarding (a) inhibits mood states such as fatigue, hunger, or/and low expectations, and (b) activates moods and cognitive states such as optimism and energy, helping to explain certain descriptive and behavioral characteristics of flow (e.g., vitality, long durations without eating). In addition, van der Linden et al. (2021) suggested that the characteristics of flow can be partially explained by the interaction between three large networks in the brain. Van der Linden et al. argued that during flow, the default mode network (DMN; as explained earlier) activity is inhibited causing absorption effects (e.g., loss of self-consciousness), the central executive network becomes activated supporting the high degree of focused attention in flow, and the salience network (that continuously evaluates feedback and affirms that one is still "in control") activating may account for the

high degree of control felt during flow and the continual inhibition of negative stress reactions. Second, Peifer and Tan (2021) published a revised and updated version of the article, "Psychophysiological Correlates of Flow Experience" (Peifer, 2012), offering an integrative account on physiological activation during flow. Peifer and Tan described findings that mirror those presented in this review (moderate sympathetic, cortisol, and parasympathetic activation, reduced DMN activity, and synchronized brain regions involving reward and cognitive control), though in addition, they also suggested a high activation of the brain's cognitive "multiple-demand system." And third, Leroy and Cheron (2020) highlighted the neural generators and rapid changes in cortical or subcortical activity that characterize flow. Specifically, Leroy and Cheron experimentally analyzed a tight-rope walker under ecological conditions and reported that the flow state was accompanied by the recruitment of additional EEG generators (globus pallidus) and basal ganglia activity (suggesting transient hypofrontality to momentarily suppress the analytical metaconscious abilities of the explicit system) not found in other high challenging (but not flow) situations.

Unanswered Questions and Future Directions

Despite providing insight into important conceptual and methodological issues, this review also highlighted key areas that require greater scrutiny by those who study flow. First, in examining the causal processes involved in flow experiences, constructs are often interchangeably referred to as antecedents, preconditions, mediators, compounding factors, or influencing factors. Moving forward, we recommend that constructs recognized or targeted as causing flow experiences to be labeled as "antecedents" so that research can be more easily amalgamated. Second, it is important that researchers' attention be directed toward understanding the nature of flow within groups. Recent research regarding flow in groups is sparse—only 6.33% of studies in this review fully or partially focused on the concept of group flow. Nevertheless, there is preliminary evidence that flow may have a contagion effect in groups (Culbertson et al., 2015; Kim, 2016). There was inconsistency among sources in this review as to whether "group flow" or "collective flow" may reflect a process of individuals being "drawn into" flow by one another in a contagion-like manner (i.e., an amalgamation of individual flow experiences; e.g., Culbertson et al., 2015), or whether a higher-order (group-level) flow construct exists (e.g., Corbato, 2018; Kim, 2016). To better understand the nature of any group-level flow construct—such as in work or sport teams—it would be valuable to draw from Chan's (1998) classic text in which he outlines the ways through which constructs (e.g., flow) may operate at different levels of analysis. The careful development and testing of composition models in the flow literature (i.e., specifying how lower- and higher-order flow constructs are related) may enable interesting research into potential group-level correlates of any higher-order construct, such as heightened social awareness (Lucas, 2017), a sense of unity or joint progress, and shared identity (van den Hout et al., 2018). In addition, such work would enable researchers to test proposals relating to theorized antecedents of flow in team settings, such as institutional, cognitive, and affective trust (Corbato, 2018), collective ambition, common goals, open communication, skill

integration, safety, mutual commitment (van den Hout et al., 2018), and collective efficacy beliefs (Salanova et al., 2014).

Aside from considering group-level, team-related issues, it is also important that researchers attend to integrating flow “theory” with other established theoretical models. It was apparent in these review sources, that there are overlaps between flow components and those in other prominent theoretical frameworks. If research efforts are directed toward studying these “overlaps” in greater detail, it is possible that such work will lead to a more nuanced understanding of (what appear to be) the common flow characteristics and antecedents outlined previously. For instance, the evidence relating to optimal challenge and arousal within flow appears to have connections to research on stress and optimal functioning (Tozman & Peifer, 2016). Specifically, the notions of “challenge” within the transactional-stress model (Lazarus, 1993; Lazarus & Folkman, 1984), “eustress” within Selye’s (1983) general adaptation syndrome, and “optimal arousal” within Hanin’s (2000) individual zones of optimal functioning model, all emphasize deep concentration and optimal functioning occurring at the top of an inverted-U bell curve (Tozman et al., 2015; Tozman & Peifer, 2016). Similarly, a closer consideration of self-efficacy theory (Bandura, 1997) may inform flow researchers’ understanding of the “optimal challenge” antecedent, and aspects of self-determination theory (SDT; Ryan & Deci, 2000) and motivational competence (see Engeser & Schiepe-Tiska, 2012) may offer insight into the “high motivation” antecedent theme. As one example, Schüller and Brandstätter (2013) reported, for instance, that subjects whose psychological (competence and relatedness) needs were not satisfied reported less flow. Lastly, the two antecedents (i.e., optimal challenge and high motivation), as posited in this review, appear to be consistent with Petty and Cacioppo’s (1986) delineation between ability and motivation as unique attributes to changing behavior.

There are also a range of important measurement and methodological considerations that stem from the findings of this review. We have already considered measurement issues relating to the assessment of flow experiences, but with respect to antecedents, it appears it would be valuable to assess optimal levels of challenge using instruments in which “demands” are not separated from “ability” (Keller & Landhausser, 2012). It would also be valuable to determine the extent to which optimal challenge is operationally distinguishable and independent from one’s high motivation, and the combinations between these factors that are most likely to give rise to intense and sustained flow experiences. Research is also needed to better understand the types of motivation that lead to flow—perhaps most notably, it would be interesting to test in more detail whether autonomous motives are necessary for flow experiences, or whether (and if so, when, and in what relation to intrinsic motives) flow may occur in the presence of more controlled motives. Finally, researchers are encouraged to continue to chart the physiological markers that characterize flow experiences, partly with an emphasis on understanding whether these processes help reveal a graduated process for entering flow (vs. a more discrete “turning on” of flow).

Practical Implications

Before considering limitations of this study and offering conclusions, it is important to note the practical implications that

emerged from this review. For those who seek to experience, train, or facilitate flow, review sources indicated that there may be promise in the use of self-regulatory training (Riva et al., 2016), hypoeu-goic self-regulation (Nakamura & Roberts, 2016), metacognition of flow training (Moneta, 2017; Wilson & Moneta, 2016), and flow mindset educational interventions (Norsworthy et al., 2017). The two broad antecedent themes outlined in this review provide direction that may help individuals achieve flow (e.g., generating high levels of motivation, and seeking challenging opportunities while maintaining a sense of capability to meet these demands and foster an optimal level of challenge). Further, Swann et al. (2017) suggested that researchers report on, or address directly, the restoration and temporal characteristics of flow experiences. Recommendations with this goal in mind might include attention training or meditation that targets focused and undistracted attention, and relaxation techniques that help manipulate individuals’ arousal levels and parasympathetic activity during arousing activities (see MacNeill & Cavanagh, 2013; Peifer et al., 2014). We additionally suggest that noninvasive electrocardiographic biofeedback devices incorporating HRV training may also prove useful in this respect, especially when technology in this space becomes mobile.

Limitations

It is worth reiterating that although our aims were suited to the conduct of a scoping review, it remains that systematic reviews and meta-analyses within more specific domains (or focused on more specific questions) of application may help further refine our knowledge of applied flow science. It may also be valuable in the future to apply a Delphi method (see Landeta, 2006) to the themes presented and conclusions offered in this review, and to solicit expert commentary from those representing the subdisciplines from which review sources were drawn. A specific limitation of our Integration of Postreview Research is that included studies were not a comprehensive representation of all flow research that has appeared in the literature between 2019 and the time of publication. Rather, in that section, we simply hoped to highlight a *selection* of what we consider to be important contemporary articles published outside the window of our review period.

In relation to the limitations of review sources, it became apparent during our analysis that more experimental studies utilizing the same conceptualization of flow (see Moving Toward Conceptual Clarity and Parsimony) are required to draw more conclusive findings. Further, authors from review sources varied in their framing of constructs as flow antecedents, preconditions, mediators, compounding factors, influencing factors, experiential descriptors, and even outcomes—making the interpretation and synthesis of research difficult. The rise of nonpsychological (i.e., neuroscientific, physiological) research is helping to further our appreciation about the nature of flow (as evident by this review), although many of the review sources (a) assumed an “optimal challenge” was the only flow antecedent, and (b) assumed that flow was apparent when difficulty levels were *objectively* balanced with skill levels. It was unclear, therefore, in certain instances, what exactly was being measured (e.g., flow or a high level of challenge non-flow state). Given that flow is a subjective phenomenon, objective measures should be analyzed alongside (rather than replacing) subjective experiential measures.

Conclusion

The work of Csikszentmihalyi and of others who have devoted sustained effort to examining and describing flow is of great importance. Flow research has been, and remains, pervasive across diverse scientific disciplines. In recent years though, scholars have noted that the language, definitions, measurement approaches, and models used in flow research and writing have become increasingly diverse. The aim of this review was to collate findings and perspectives on flow from different scientific disciplines. In doing so, we sought to identify what was common across recent flow work, and to make a step toward finding a more consistent language and meaning for those who work to understand and promote flow experiences. The findings of this review indicate that despite substantial diversity in terminology, concepts, descriptors, and assessment methods, there are indeed many commonalities in our collective study of this construct. Focusing on these commonalities may advance our understanding of flow beyond the nine-dimensional model and better inform us about what flow is, what flow is not, and how flow is generated and best measured. Moreover, the systematic study of these commonalities promises to aid our conceptual understanding, help address measurement limitations, enable better theoretical integration, and may allow people to more readily facilitate and experience flow.

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